CHAPTER 2

REPRODUCTIVE HEALTH

CHAPTER ESSENCE AT A GLANCE:
Contraceptive intervention strategies, STI prevention, and assisted reproductive technologies promoting reproductive wellness.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Reproductive Health Programs and Amniocentesis:** India pioneered national family planning programs in 1951, modernized as Reproductive and Child Health Care (RCH) programs. Amniocentesis analyzes fetal cells in amniotic fluid for chromosomal abnormalities (e.g., Down's syndrome); a statutory ban on its use for prenatal sex determination was enacted in India to combat female foeticide and sex-ratio imbalance.

• **Contraceptive Strategies and Mechanism Classification:** Contraceptive methods: Natural (periodic abstinence during fertile window days 10 to 17, coitus interruptus, lactational amenorrhea up to 6 months); Barrier methods (condoms preventing STIs, diaphragms); Intrauterine Devices (IUDs): Non-medicated (Lippes loop), Copper-releasing (CuT, Cu7, Multiload 375: Cu ions suppress sperm motility and fertilizing ability), and Hormone-releasing (Progestasert, LNG-20: make uterus unsuitable for implantation and cervix hostile to sperm); Oral pills (Saheli: non-steroidal once-a-week pill developed by CDRI Lucknow); Surgical sterilization (Vasectomy in males, Tubectomy in females: permanent blocking of gamete transport).

• **Medical Termination of Pregnancy (MTP) and STIs:** Medical Termination of Pregnancy (MTP) is legal under the MTP Amendment Act 2017 up to 12 weeks (first trimester safe window), permitted to prevent unwanted pregnancies or maternal or fetal mortality risks. Sexually Transmitted Infections (STIs): Gonorrhea, Syphilis, Genital herpes, Chlamydiasis, Hepatitis-B, and HIV; complications of untreated STIs include Pelvic Inflammatory Disease (PID), ectopic pregnancies, and infertility.

• **Assisted Reproductive Technologies (ART):** ART addresses human infertility: In Vitro Fertilization (IVF) followed by Embryo Transfer (ET); ZIFT (Zygote Intra-Fallopian Transfer: zygote or embryo up to 8 blastomeres transferred into fallopian tube); IUT (Intra-Uterine Transfer: embryo with more than 8 blastomeres transferred into uterus); GIFT (Gamete Intra-Fallopian Transfer: transfer of ovum from donor into fallopian tube of female who cannot produce eggs); ICSI (Intra-Cytoplasmic Sperm Injection: sperm directly injected into ovum in laboratory); IUI (Intra-Uterine Insemination: semen introduced into uterus).

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$\text{IUD Taxonomy} = \frac{\text{Copper: CuT, Multiload 375 (Suppress sperm motility)}}{\text{Hormone: LNG-20, Progestasert (Hostile endometrium/cervix)}}$$

Mode of Action Classification for Intrauterine Devices

$$\text{Embryo Transfer Modality} = \frac{\text{ZIFT: } \leq 8 \text{ blastomeres into Fallopian tube}}{\text{IUT: } > 8 \text{ blastomeres directly into Uterus}}$$

Blastomere Cleavage Thresholds in ART Embryo Transfer

$$\text{Surgical Sterilization} = \text{Vasectomy (Vas deferens excised)} \Longleftrightarrow \text{Tubectomy (Fallopian tubes excised)}$$

Irreversible Surgical Gamete Barrier Procedures

$$\text{Oral Contraceptive Innovation} = \text{Saheli (Centchroman by CDRI)} \rightarrow \text{Non-steroidal once-a-week pill}$$

Non-Steroidal Selective Estrogen Receptor Modulator

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In ZIFT, embryos up to 8 blastomeres are transferred into the fallopian tube; in IUT, embryos with more than 8 blastomeres are transferred into the uterus.

• **Alert:** Copper ions released from Cu-releasing IUDs suppress sperm motility and the fertilizing capacity of spermatozoa.

• **Alert:** Saheli is a non-steroidal oral contraceptive pill taken once a week with very few side effects, developed by CDRI in Lucknow.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is amniocentesis? Why is there a statutory ban on its use for prenatal sex determination in India? [Score: 2 Marks]

Answer & Key Valuation Steps:

• Amniocentesis: A prenatal diagnostic technique where amniotic fluid containing fetal cells is sampled to detect genetic or chromosomal disorders (e.g., Down's syndrome, sickle-cell anemia). **[1 Mark]**

• Statutory ban justification: Banned to prevent its illegal misuse for prenatal sex determination, which leads to female foeticide and distorted sex ratios. **[1 Mark]**

2. Distinguish between copper-releasing IUDs (e.g., CuT) and hormone-releasing IUDs (e.g., LNG-20) based on their contraceptive mechanisms. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Copper-releasing IUDs (CuT, Multiload 375): Release Cu ions that suppress sperm motility and reduce the fertilizing capacity of spermatozoa in the uterine cavity. **[1 Mark]**

• Hormone-releasing IUDs (Progestasert, LNG-20): Release progestins that make the uterus unsuitable for implantation and render the cervix hostile and impenetrable to sperm. **[1 Mark]**

3. Classify surgical methods of contraception in human males and females, describing vasectomy and tubectomy. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Surgical sterilization methods block gamete transport permanently, providing highly effective contraception with poor reversibility. **[1 Mark]**

• Vasectomy (males): A small part of the vas deferens is removed or tied up through a small incision on the scrotum, preventing sperms from reaching the semen. **[1 Mark]**

• Tubectomy (females): A small part of both fallopian tubes is removed or tied up through a small incision in the abdomen or vagina, preventing ovum from meeting sperm. **[1 Mark]**

4. Explain the following Assisted Reproductive Technologies (ART): (a) ZIFT, (b) GIFT, and (c) ICSI, indicating the clinical condition for each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• (a) ZIFT (Zygote Intra-Fallopian Transfer): Zygote or early embryo up to 8 blastomeres formed in vitro is transferred into the fallopian tube of an infertile woman with patent tubes. **[1 Mark]**

• (b) GIFT (Gamete Intra-Fallopian Transfer): Transfer of an ovum collected from a donor female into the fallopian tube of another female who cannot produce an ovum but provides suitable environment for fertilization. **[1 Mark]**

• (c) ICSI (Intra-Cytoplasmic Sperm Injection): A laboratory procedure where a single sperm is directly injected into the cytoplasm of an ovum to form an embryo, used in cases of severe oligospermia. **[1 Mark]**

5. What is Medical Termination of Pregnancy (MTP)? Up to which stage of pregnancy is MTP legally permitted and considered medically safe in India? [Score: 3 Marks]

Answer & Key Valuation Steps:

• Definition: Intentional or voluntary termination of pregnancy before full term, also termed induced abortion. **[1 Mark]**

• Safe window: MTP is considered relatively safe during the first trimester (up to 12 weeks of pregnancy); second trimester terminations are much riskier. **[1 Mark]**

• Legal status: Legalized in India in 1971 with strict conditions to avoid misuse; under the MTP Amendment Act 2017, termination up to 12 weeks requires the opinion of one registered medical practitioner. **[1 Mark]**

6. List four common Sexually Transmitted Infections (STIs). State two major medical complications that can arise if STIs are left undiagnosed and untreated. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Four common STIs: (1) Gonorrhea, (2) Syphilis, (3) Genital herpes, (4) Chlamydiasis (or Hepatitis-B, HIV). **[1.5 Marks]**

• Complications of untreated STIs: Can lead to Pelvic Inflammatory Disease (PID), abortions, stillbirths, ectopic pregnancies, and irreversible infertility or pelvic cancer. **[1.5 Marks]**

7. A married couple has been unable to conceive a child for three years. Clinical investigation reveals that the wife has permanently blocked fallopian tubes but ovulates normally, and the husband has a normal sperm count. Suggest the most appropriate ART procedure for this couple, and explain the step-by-step laboratory and clinical protocol. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Recommended ART procedure: In Vitro Fertilization followed by Embryo Transfer (IVF-ET). **[1 Mark]**

• Oocyte and sperm harvesting: Ova are retrieved from the wife's ovaries under ultrasound guidance after hormonal superovulation; fresh motile sperms are collected from the husband. **[1 Mark]**

• In vitro fertilization: Ova and sperms are co-incubated in a culture dish under sterile laboratory conditions mimicking physiological conditions to permit fertilization and cleavage. **[1 Mark]**

• Clinical embryo transfer: Because fallopian tubes are blocked, transfer cannot be ZIFT; instead, embryos cultured past the 8-cell stage (morula or blastocyst) are transferred directly into the uterine cavity via Intra-Uterine Transfer (IUT) for endometrial implantation. **[1 Mark]**